

Lab Sheet 04

1. One-Sample t-Test

- To determine if the mean of a single sample differs from a specified value

Create vector to hold plant heights

```
my_data = c(14, 14, 16, 13, 12, 17, 15, 14, 15, 13, 15, 14)
```

Perform one-sample t-test

```
t.test(my_data, mu = 15)
```

- mu - The hypothesized mean value
 - Output includes the t-statistic, degrees of freedom, p-value, and confidence interval
-

2. Two-Sample t-Test

- To compare the means of two independent samples

Create vectors for plant heights from two groups

```
group1 = c(8, 8, 9, 9, 9, 11, 12, 13, 13, 14, 15, 19)
```

```
group2 = c(11, 12, 13, 13, 14, 14, 14, 15, 16, 18, 18, 19)
```

Perform two-sample t-test

```
t.test(group1, group2, var.equal = TRUE)
```

- var.equal - Specifies whether to assume equal variances
- Output includes comparison of means, confidence intervals, and p-value

3. One-Sample z-Test

- To test if the mean of a sample differs from a known population mean when the population standard deviation is known

Load BSDA library

```
library(BSDA)
```

IQ levels for 20 patients

```
data = c(88, 92, 94, 94, 96, 97, 97, 97, 99, 99, 105, 109, 109, 109, 110, 112, 112, 113, 114, 115)
```

Perform one-sample z-test

```
z.test(data, mu = 100, sigma.x = 15)
```

- mu - Population mean
- sigma.x - Known standard deviation

4. Two-Sample z-Test

- To compare the means of two independent samples with known population standard deviations

IQ levels for individuals in two cities

```
cityA = c(82, 84, 85, 89, 91, 91, 92, 94, 99, 99, 105, 109, 109, 109, 110, 112, 112, 113, 114, 114)
```

```
cityB = c(90, 91, 91, 91, 95, 95, 99, 99, 108, 109, 109, 114, 115, 116, 117, 117, 128, 129, 130, 133)
```

Perform two-sample z-test

```
z.test(x = cityA, y = cityB, mu = 0, sigma.x = 15, sigma.y = 15)
```

5. Variance Ratio Test (F-Test)

- To compare variances of two independent samples

Plant heights from two groups

```
group1 = c(5, 6, 6, 8, 10, 12, 12, 13, 14, 15, 15, 17, 18, 18, 19)
```

```
group2 = c(9, 9, 10, 12, 12, 13, 14, 16, 16, 19, 22, 24, 26, 29, 29)
```

Perform variance ratio test

```
var.test(group1, group2)
```

6. One-Proportion z-Test

- To test if a population proportion differs from a specified value

Proportion of COVID-positive patients

```
prop.test(x = 60, n = 100, p = 0.5, alternative = "greater")
```

- x - Number of successes
 - n - Sample size
 - p - Hypothesized proportion
-

7. Two-Proportions z-Test

- To compare proportions between two groups

Proportions of COVID-positive patients in two groups

```
prop.test(x = c(60, 80), n = c(100, 200), alternative = "greater")
```

8. Analysis of Variance (ANOVA)

- To compare means across multiple groups

```
# Load iris dataset
```

```
data("iris")
```

```
colnames(iris) = c("SL", "SW", "PL", "PW", "Specs")
```

```
st_data = stack(iris[, 1:4])
```

```
# Perform ANOVA
```

```
ano_tab = aov(values ~ ind, data = st_data)
```

```
summary(ano_tab)
```

9. Levene's Test

- To test equality of variances

```
# Load car library
```

```
library(car)
```

```
# Perform Levene's test
```

```
leveneTest(values ~ ind, data = st_data)
```

10. Post-Hoc Analysis - Fisher's LSD

- To perform pairwise comparisons after ANOVA

```
# Create data frame

df = data.frame(technique = rep(c("tech1", "tech2", "tech3"), each = 10),
               score = c(72, 73, 73, 77, 82, 82, 83, 84, 85, 89,
                        81, 82, 83, 83, 83, 84, 87, 90, 92, 93,
                        77, 78, 79, 88, 89, 90, 91, 95, 95, 98))

# Fit one-way ANOVA

model = aov(score ~ technique, data = df)

summary(model)

# Fisher's LSD

library(agricolae)

LSD.test(model, "technique")
```

11. Testing Normality

- To check if data follows a normal distribution

```
# Generate random data from normal distribution

data = rnorm(100)

shapiro.test(data)

hist(data, col = "steelblue")

# Check normality for iris dataset

shapiro.test(iris$Sepal.Length)
```

```
hist(iris$Sepal.Length)
```

12. Correlation Analysis

- To measure the strength and direction of relationships between variables

```
# Data frame
```

```
df = data.frame(hours = c(1, 1, 3, 2, 4, 3, 5, 6),  
                prac_exams = c(4, 3, 3, 2, 3, 2, 1, 4),  
                score = c(69, 74, 74, 70, 89, 85, 99, 90))
```

```
# Pearson correlation
```

```
cor(df$hours, df$score)
```

```
cor(df)
```

```
# Correlation test
```

```
x = c(2, 3, 3, 5, 6, 9, 14, 15, 19, 21, 22, 23)
```

```
y = c(23, 24, 24, 23, 17, 28, 38, 34, 35, 39, 41, 43)
```

```
# Scatterplot
```

```
plot(x, y, pch = 16, col = "red")
```

```
cor.test(x, y)
```